

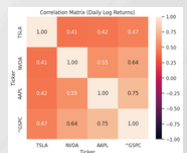


The Tech Domino Effect: Volatility Transmission from Mega-Caps to the S&P 500

Vo Thi Quynh Tram, Lai Chiu Lok, Huei-Wen Teng

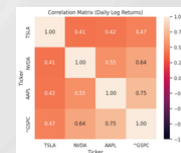
Motivation& Introduction

- Rising dominance of mega-cap tech firms in the market.
- Potential volatility spillovers affecting overall market movements.
- Need to assess systemic risk from concentrated market influence.



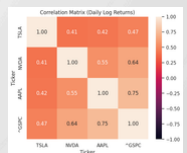
Outline

1. Motivation & Introduction ✓
2. Data & Benchmark method
3. Study plan & results



Data Preprocess_Data Overview

- ▣ Source : Yahoo Finance (Adjusted Close)
- ▣ Period : 2015–2024 (Daily)
- ▣ Target Assets : Apple (AAPL), Nvidia (NVDA), Tesla (TSLA),
S&P 500 (Benchmark)
- ▣ Transformation : Log returns → Rolling Volatility (annualized)



Data Preprocess_Variables

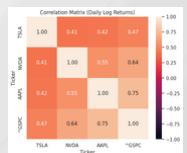
Variable	Definition	Role
RV_SPX	Rolling Volatility of S&P 500	Dependent
RET_AAPL, RET_NVDA, RET_TSLA	Log Returns of tech firms	Independent
RV_AAPL, RV_NVDA, RV_TSLA	Rolling Vol of tech firms	Spillover Drivers



Data Preprocess_Descriptive Statistics

Ticker	count	mean	std	min	25%	50%	75%	max	Skew	Kurtosis	JB_pvalue
AAPL	2514	0.000930	0.017934	-0.137708	-0.007355	0.001000	0.010093	0.113158	-0.201228	5.457399	0
NVDA	2514	0.002248	0.030378	-0.207712	-0.012514	0.002648	0.017472	0.260876	0.208734	6.829886	0
TSLA	2514	0.001333	0.035911	-0.236518	-0.016292	0.001264	0.019073	0.198187	-0.066409	4.410539	0
SPX	2514	0.000419	0.011272	-0.127652	-0.003780	0.000639	0.005740	0.089683	-0.810830	15.750819	0

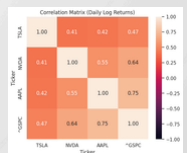
- ▣ All assets: high variance (Std highest for TSLA)
- ▣ Evidence of non-normality:
 - ▶ Positive/negative skew → asymmetric risk
 - ▶ High kurtosis → heavy tails
 - ▶ JB-test → rejects normality ($p = 0$)



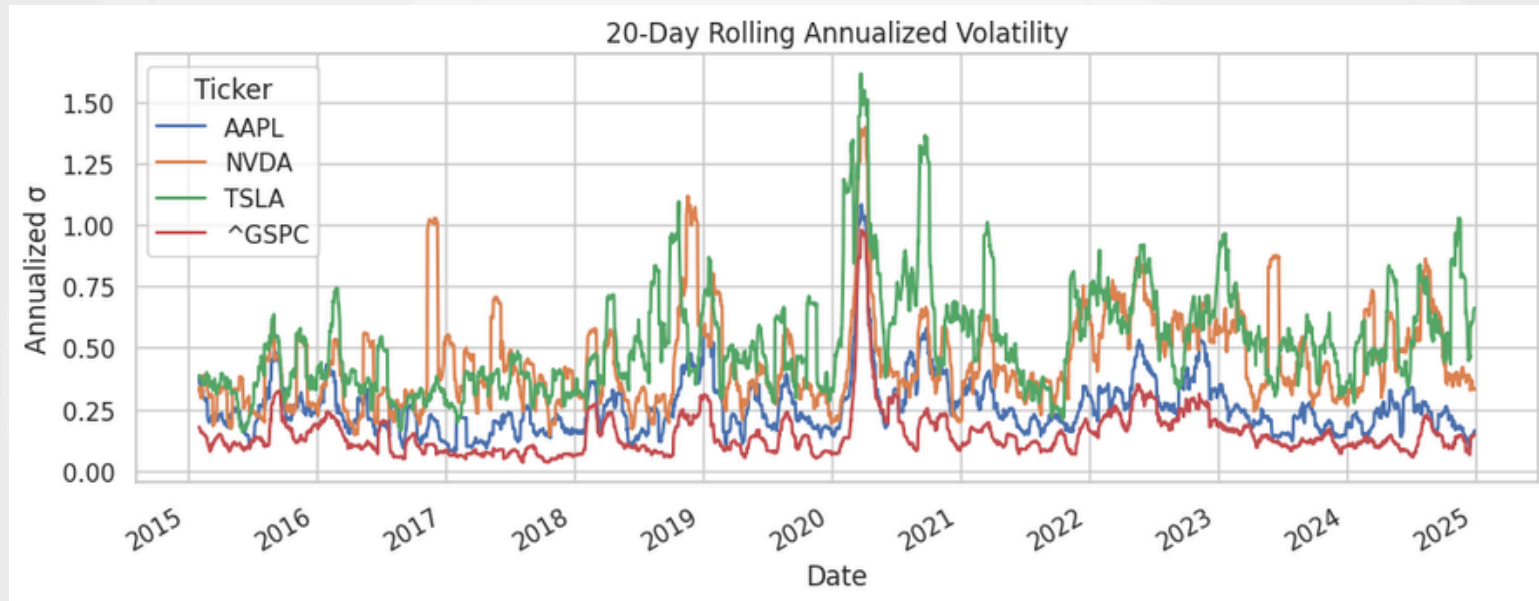
Data Preprocess_Extreme Events ($|z| > 3$)

Asset	Count
TSLA	40
AAPL	40
NVDA	29
S&P500	36

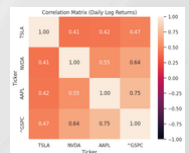
TSLA & AAPL most prone to tail shocks → candidates for volatility domino effects.



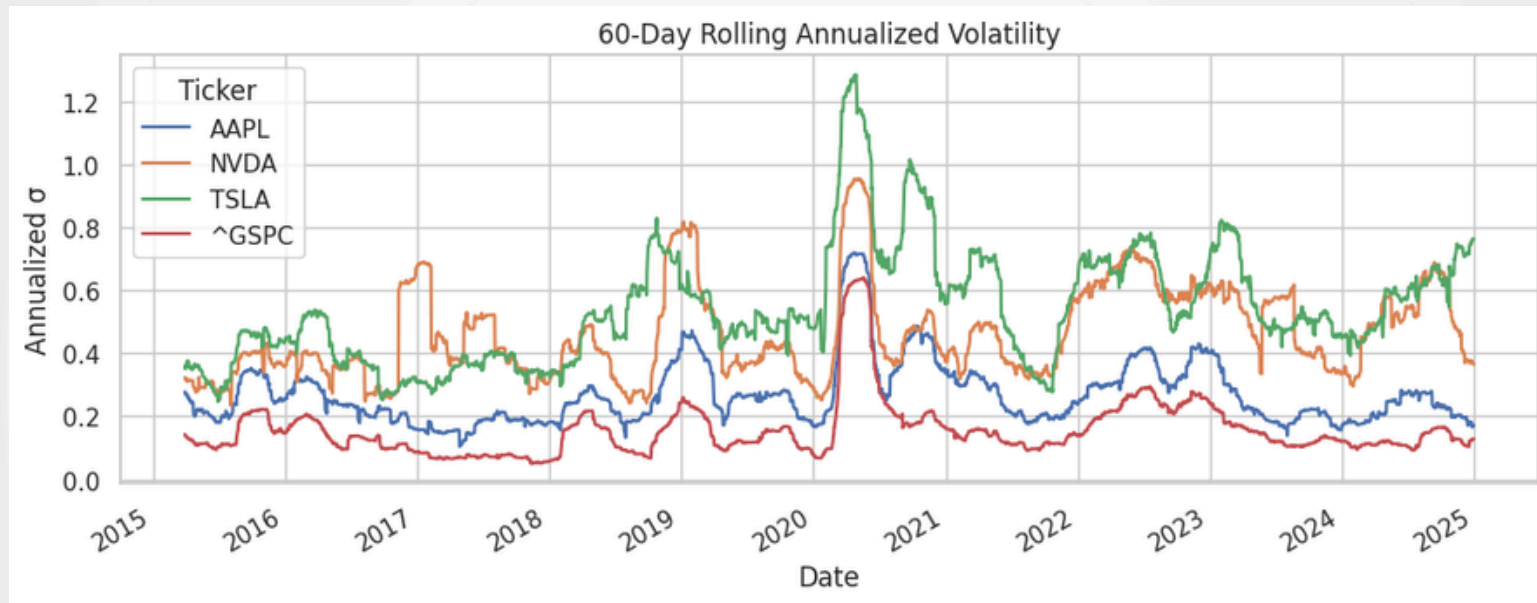
Data Preprocess_Rolling Volatility



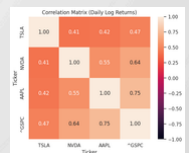
- TSLA & NVDA show sharp volatility spikes (2020–2021)
- SPX volatility rises after mega-cap spikes → Possible contagion
- AAPL = moderate, smoothed volatility (market stabilizer)



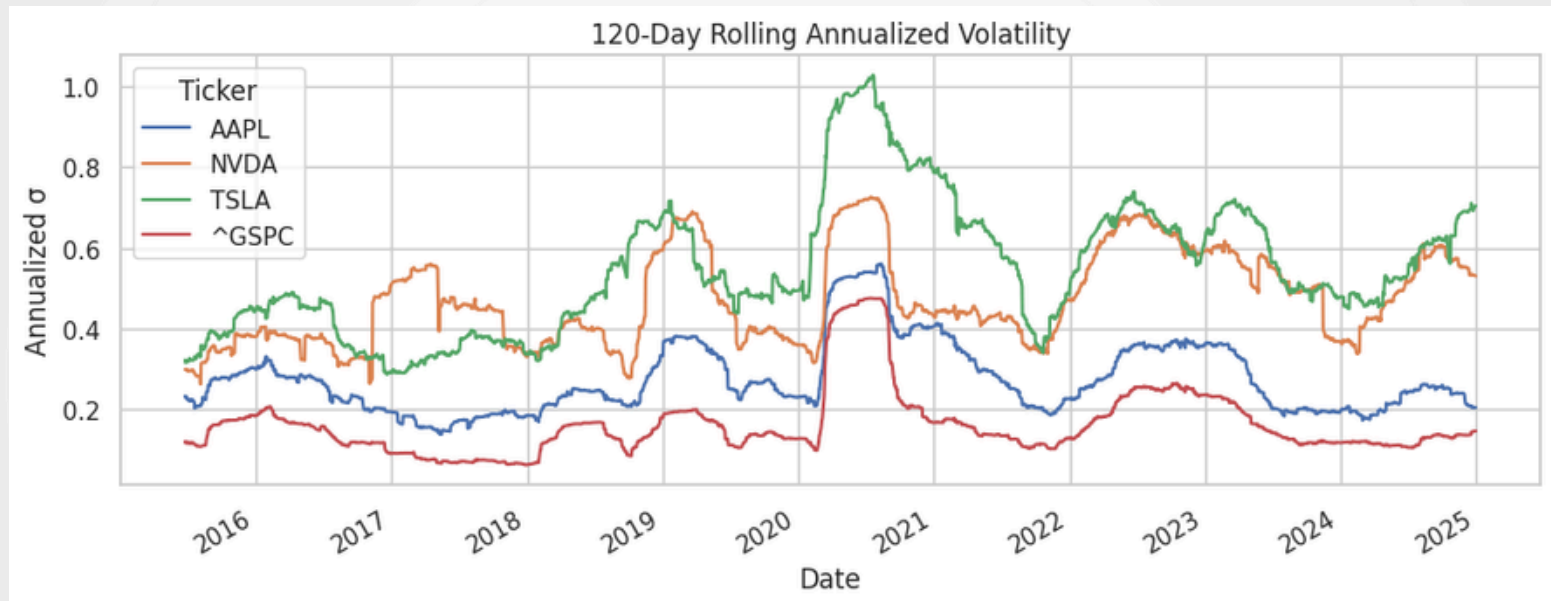
Data Preprocess_Rolling Volatility



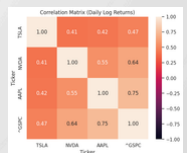
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Data Preprocess_Rolling Volatility



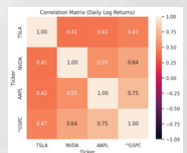
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Data Preprocess_Market Sensitivity

Asset	Beta vs S&P500
NVDA	1.73
TSLA	1.5
AAPL	1.19

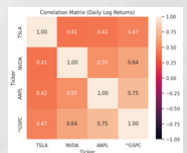
- ▣ NVDA & TSLA amplify market risk (“leveraged exposure”)
- ▣ High Beta supports volatility transmission hypothesis



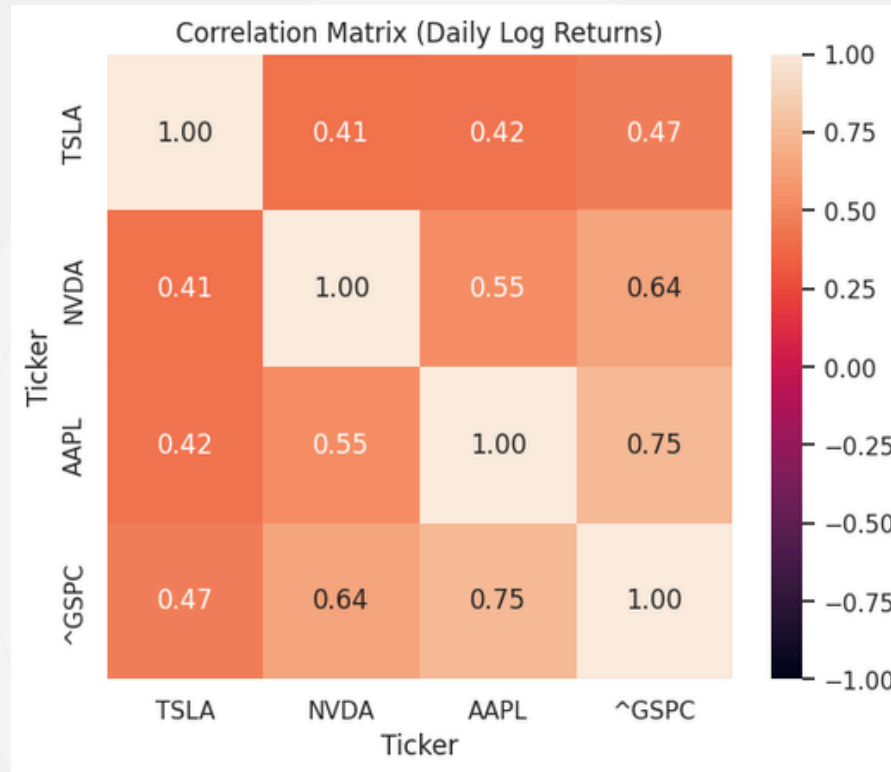
Data Preprocess_Risk Metrics (MaxDD,CVaR)

Asset	MaxDD	CVaR(95%)
TSLA	-73.6%	-0.083
AAPL	-66.3%	-0.067
NVDA	-38.5%	-0.042
S&P500	-33.9%	-0.028

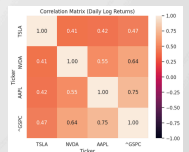
Mega-caps have deeper crash risk → Potential systemic propagation.



Data Preprocess_Correlation & Co-Movement

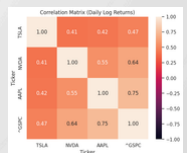


- AAPL highly correlated with SPX (~0.75)
- TSLA shows episodic decoupling (idiosyncratic risk)
- NVDA moderately correlated, rising in AI boom
- Conclusion: Strong integration but not perfect coupling → transmission, not redundancy.



Data Preprocess_Key Findings (Volatility Transmission)

- Evidence of fat tails, extreme risk, and high Beta in mega-caps
- S&P500 volatility spikes lag mega-cap volatility surges
- Tech sector acts as a volatility leader, not follower

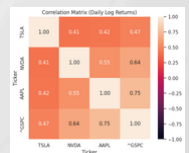
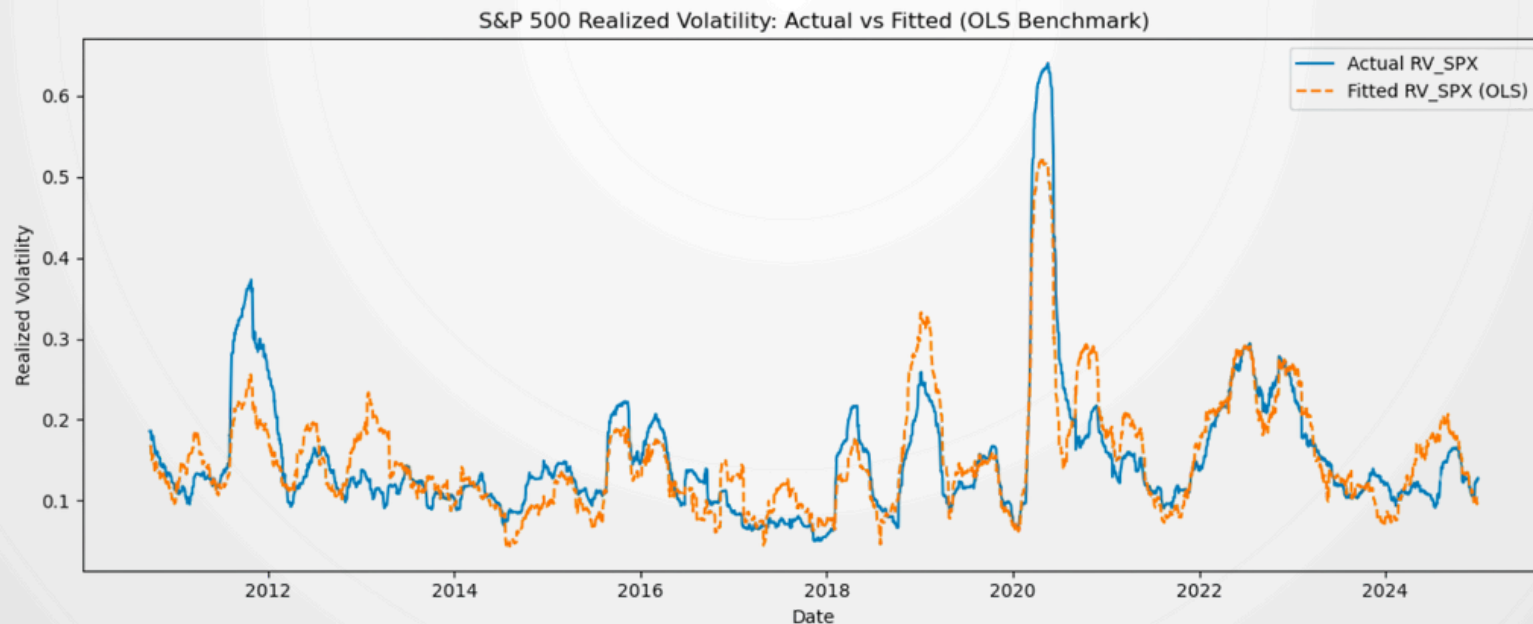


Benchmark model: OLS

The baseline regression model

$$RV_t^{SPX} = \alpha + \beta_1 RV_t^{AAPL} + \beta_2 RV_t^{NVDA} + \beta_3 RV_t^{TSLA} + \varepsilon_t,$$

where RV_t^{SPX} denotes the 60-day realized volatility of the S&P 500 and RV_t^{AAPL} , RV_t^{NVDA} , and RV_t^{TSLA} are the corresponding realizations for Apple, Nvidia, and Tesla.



Benchmark model: OLS

The baseline regression model

$$RV_t^{SPX} = \alpha + \beta_1 RV_t^{AAPL} + \beta_2 RV_t^{NVDA} + \beta_3 RV_t^{TSLA} + \varepsilon_t,$$

- The model explaining **75%** of total variation.
- **Apple and Nvidia** exhibit the largest and most statistically significant transmission effects.
- **Tesla's effect**, while significant, is markedly smaller

===== OLS BENCHMARK RESULTS =====

OLS Regression Results

Dep. Variable:

RV_SPX

R-squared:

0.750

Model:

OLS

Adj. R-squared:

0.750

Method:

Least Squares

F-statistic:

3591.

Date:

Sat, 06 Dec 2025

Prob (F-statistic):

0.00

Time:

14:04:23

Log-Likelihood:

6336.3

No. Observations:

3591

AIC:

-1.266e+04

Df Residuals:

3587

BIC:

-1.264e+04

Df Model:

3

Covariance Type:

nonrobust

coef

std err

t

P>|t|

[0.025

0.975]

const

-0.0971

0.003

-36.352

0.000

-0.102

-0.092

RV_AAPL

0.5295

0.010

51.261

0.000

0.509

0.550

RV_NVDA

0.1879

0.006

32.301

0.000

0.176

0.199

RV_TSLA

0.0479

0.005

9.076

0.000

0.038

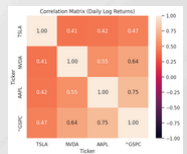
0.058

→ Provide clear evidence that a small cluster of mega-cap technology firms serves as **dominant drivers of S&P 500 volatility**

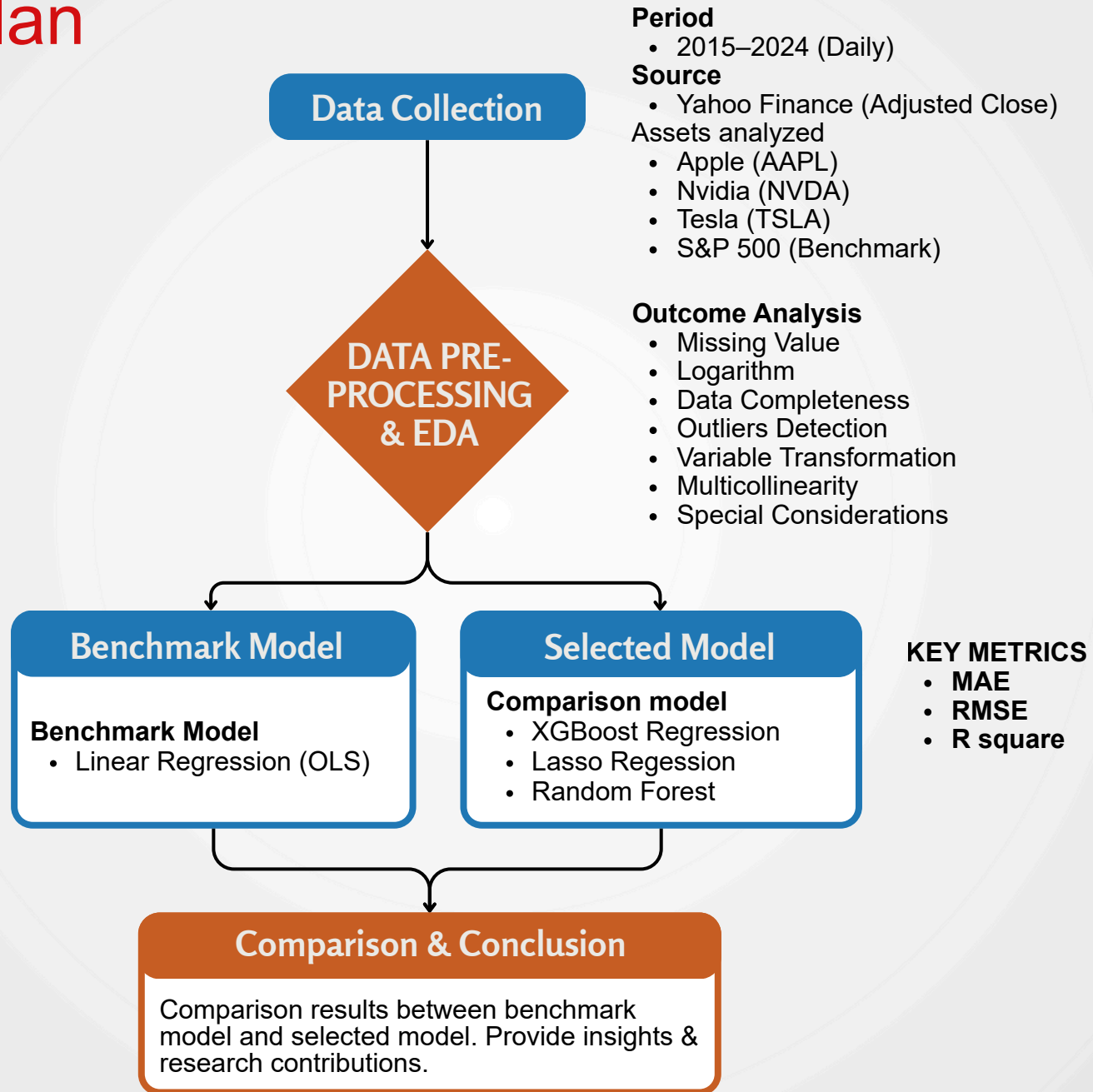


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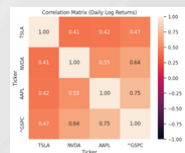
Study Plan



Selecting Models

Objective: Exhibit nonlinearity, multicollinearity, and high-dimensional interactions that OLS cannot fully capture

Model	Key Strength	Rationale for Volatility Spillover Analysis
LASSO	Feature selection + multicollinearity control	Identifies core spillover transmitters; stabilizes coefficients among highly correlated mega-caps
Random Forest	Nonlinear modeling + interaction capture	Detects nonlinear volatility spillovers and structural breaks in tech stocks
XGBoost	Highest predictive performance in finance	Learns complex crisis-sensitive and asymmetric transmission patterns ; handles outliers well



Selecting Models

XG BOOST

The XGBoost prediction is

$$\hat{y}_t = \sum_{m=1}^M f_m(\mathbf{x}_t) = \sum_{m=1}^M f_m(RV_t^{AAPL}, RV_t^{NVDA}, RV_t^{TSLA}).$$

The trees $\{f_m\}$ are obtained by minimizing the regularized objective

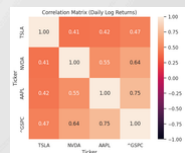
$$\mathcal{L} = \sum_{t=1}^n \ell(y_t, \hat{y}_t) + \sum_{m=1}^M \Omega(f_m),$$

where $\ell(\cdot)$ is typically the squared loss

$$\ell(y_t, \hat{y}_t) = \frac{1}{2}(y_t - \hat{y}_t)^2,$$

and $\Omega(f_m)$ is the tree regularization term (penalizing number of leaves and leaf weights)

Model	RMSE	MAE	R^2
OLS (Linear Regression)	0.032018	0.025008	0.709753
XGBoost	0.041913	0.032884	0.502621



Selecting Models

Random Forest

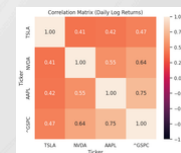
Let $T_b(\mathbf{x}_t)$ be the prediction from the b -th tree ($b = 1, \dots, B$).

The Random Forest prediction for day t is

$$\hat{y}_t = \frac{1}{B} \sum_{b=1}^B T_b(\mathbf{x}_t) = \frac{1}{B} \sum_{b=1}^B T_b(RV_t^{AAPL}, RV_t^{NVDA}, RV_t^{TSLA}),$$

where each T_b is trained on a bootstrap sample of the data and at each split only a random subset of predictors is considered

Model	RMSE	MAE	R^2
OLS (Linear Regression)	0.032018	0.025008	0.709753
Random Forest	0.048793	0.037036	0.325929



Selecting Models

LASSO

Prediction has the **same linear form** as OLS:

$$\hat{y}_t = \hat{\alpha} + \hat{\beta}_1 RV_t^{AAPL} + \hat{\beta}_2 RV_t^{NVDA} + \hat{\beta}_3 RV_t^{TSLA}.$$

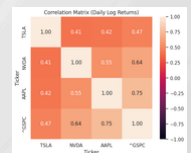
The difference is **how** $(\hat{\alpha}, \hat{\beta})$ are estimated.

LASSO solves the penalized least-squares problem (what you did with `LassoCV` in Python):

$$(\hat{\alpha}, \hat{\beta}) = \arg \min_{\alpha, \beta} \left\{ \frac{1}{2n} \sum_{t=1}^n (y_t - \alpha - \beta^\top \mathbf{x}_t)^2 + \lambda \sum_{j=1}^3 |\beta_j| \right\},$$

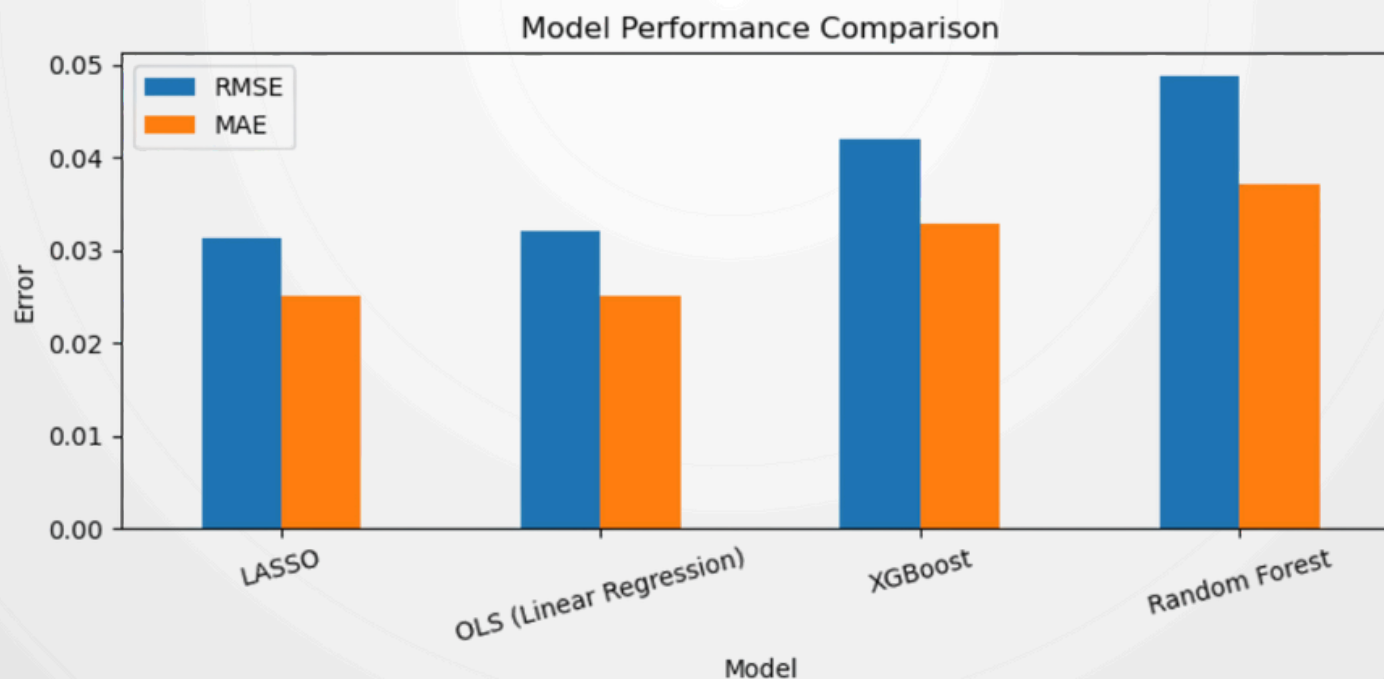
where $\lambda \geq 0$ is the regularization parameter chosen by cross-validation.

Model	RMSE	MAE	R^2
OLS (Linear Regression)	0.032018	0.025008	0.709753
LASSO	0.031233	0.025042	0.723805

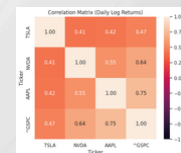


Machine-learning Model Performance

Model	RMSE	MAE	R^2
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XGBoost	0.041913	0.032884	0.502621
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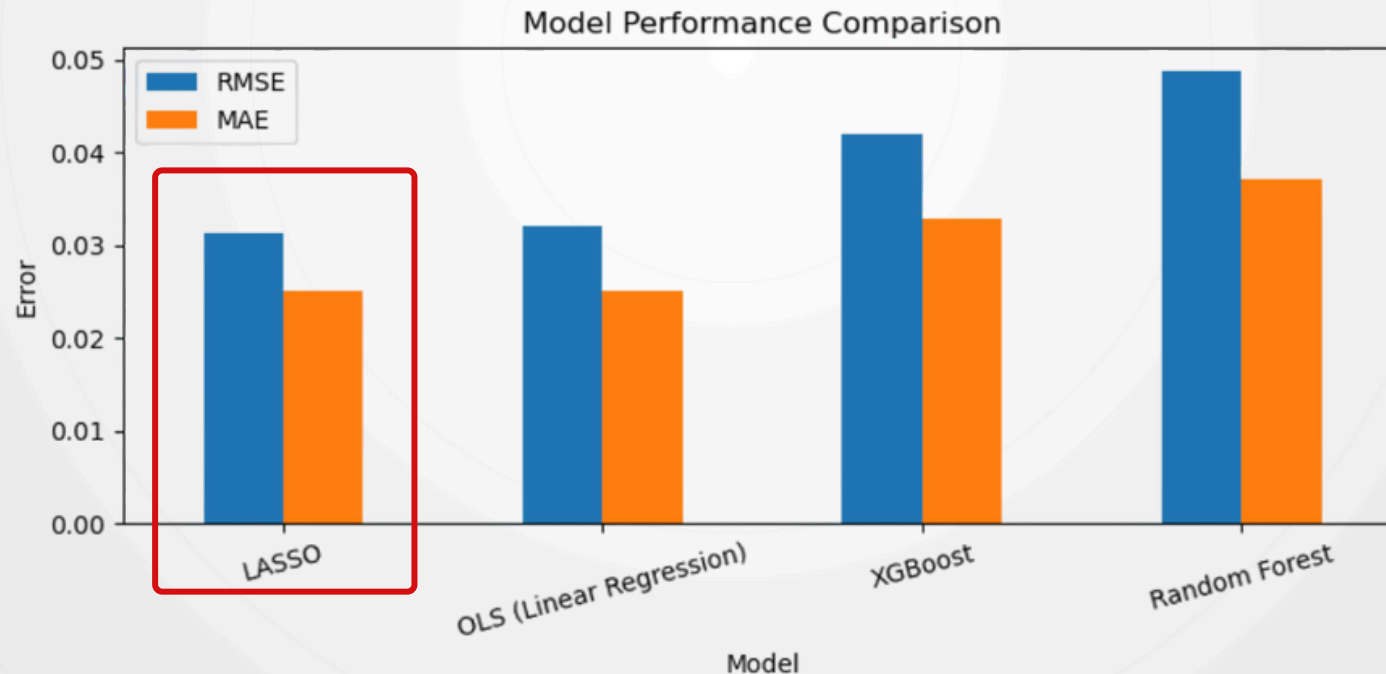
Linear Regression



Machine-learning Model Performance

LASSO achieves a better bias–variance tradeoff

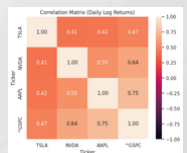
- A lower RMSE → Better predictive accuracy
- A higher R square → more variance in S&P 500 volatility.
- Comparable MAE to OLS → no loss in average prediction error



Key insight & Contribution

- **Apple's** realized volatility exhibits the **strongest spillover effect**.
- Followed by **Nvidia**, while **Tesla's** influence is positive but **comparatively modest**, indicating that its volatility is more idiosyncratic.

Small set of large technology firms plays a disproportionately systemic role in shaping aggregate U.S. market risk



Thank you

